



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY SONEPAT

भारतीय सूचना प्रौद्योगिकी संस्थान सोनीपत

(An Autonomous Institute of National Importance under Act of Parliament)

I-Tech, Techno Park, ITD Rajiv Gandhi Education City, Rai, Sonepat-131029 (Haryana)

Phone: +91 1302987921, Email: sonepatiit@gmail.com, website: www.iiitsonepat.ac.in

Mid Semester Examination October-2025

Subject: HSS (SAP 102)

Semester: 1st

Roll no

Branch: CSE-IT-DSA

M.M. -30 Marks

Time: 02 Hrs.

Note: All questions are compulsory

All questions carry equal marks

1.	What do you understand by physical fitness? With the help of a diagram, explain the health-related physical fitness components and skill related physical fitness components.	(5 marks)
2.	Define what do you understand by first aid? With the help of suitable examples, discuss the aims and objectives of first aid. W, LIR	(5 marks)
3.	Explain the importance of first aid in real life situation. Discuss various basic first aid techniques and their significance in emergency situations.	(5 marks)
4.	What do you understand by CPR. Explain various steps of CPR with the help of a diagram.	(5 marks)
5.	Discuss the definition of yoga. With the help of a diagram, illustrate various components of yoga.	(5 marks)
6.	Write a detailed note on the various types of Asanas and Pranayama. Also discuss their significance in terms of inner peace and healthy life style.	(5 marks)

(Total pages: 2)

Roll No.

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INDIAN INSTITUTE OF INFORMATION TECHNOLOGY SONEPAT
Email: sonapatiit@gmail.com, website: www.iitsonapat.ac.in

Analog Electronics (ECL106)
Mid Semester Examination (October 31, 2025)

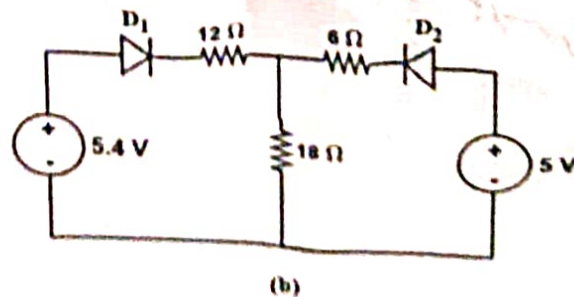
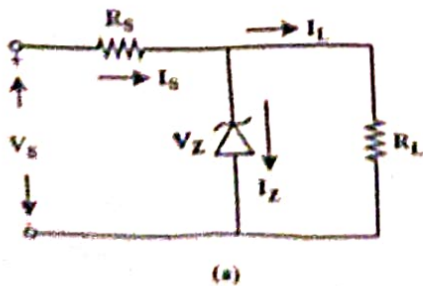
Branch: CSE/IT/DSA

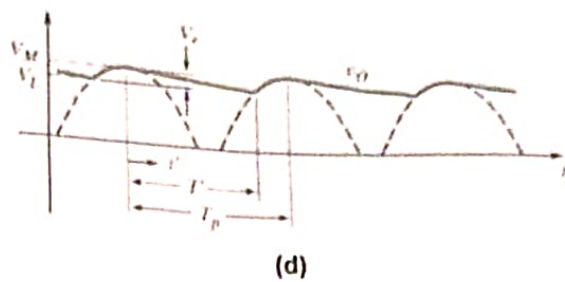
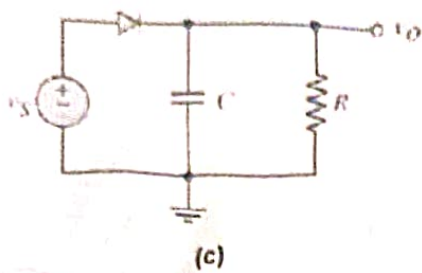
Semester- I

Max. Marks: 30

Note: All questions are compulsory

1. (a) Define the following terms of P-N junction diode: (3)
 - i) Depletion region and Barrier potential.
 - ii) Drift and Diffusion currents.
 - iii) Peak Inverse Voltage (PIV).
- (b) Name the breakdown mechanism in a heavily doped P-N junction under reverse biased condition and define it. (2)
2. (a) For the circuit shown in **Figure-(a)**, $R_S = 20 \Omega$, $V_Z = 18 \text{ V}$ and $R_L = 200 \Omega$. If V_S varies from 20 V to 30 V, find (3)
 - i) The minimum and maximum currents through Zener diode.
 - ii) The minimum and maximum power dissipated in the Zener diode.
 - iii) The maximum rated power dissipation that R_S should have.
- (b) In the circuit shown in **Figure-(b)**, the diodes has cut-in voltage of 0.7 V. What is the status of each diode? Also find the values of currents through diodes D_1 and D_2 . (2)
3. (a) Prove that the rectification efficiency η of bridge full wave rectifier circuit having ideal diodes is 81.2 %. (3)





- (b) The output (v_s) of a center-tapped full wave rectifier is applied to $R-C$ filter circuit is shown in **Figure-(c)**. The output of that circuit is shown in **Figure-(d)**. Prove that the ripple voltage (V_r) is equal to $V_M/2fRC$, where the frequency (f) is the ac input frequency applied to rectifier circuit. (2)

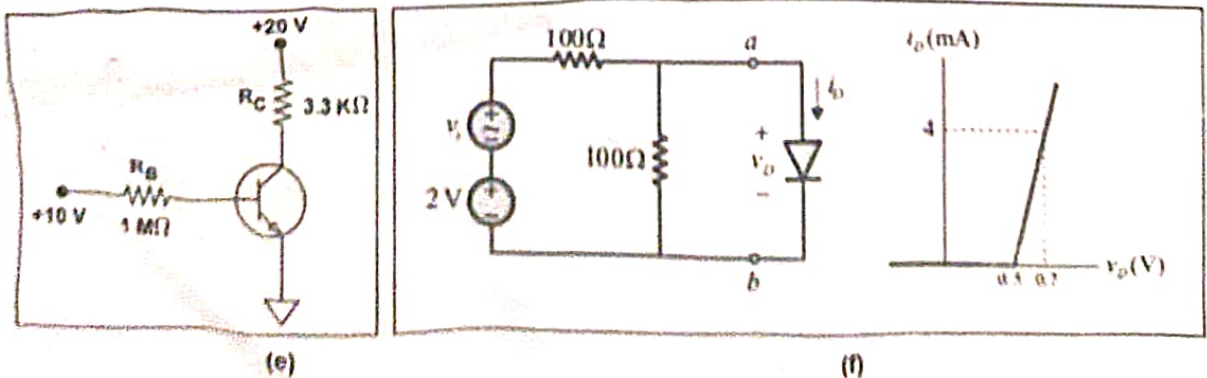
4. (a) Define the following terms related to Bipolar Junction Transistor (BJT) (3)
- Operating point and Stability factor.
 - Why is the size of collector region kept maximum in BJT?

- (b) The BJT has emitter current $I_E = 10$ mA and $\alpha = 0.98$. Determine the values of I_B and β . (2)

5. (a) Define a Transistor and its applications. Explain the working of n-p-n transistor for common emitter configuration. Also draw its input and output characteristics. *C&E* *B-7FB* (3)

- (b) Draw the DC load line for the circuit as shown in **Figure-(e)**. What is the region of operation of silicon transistor having $\beta = 100$? Find the values of V_{CE} and I_C . (2)

- (c) The diodes in the circuit has the non-linear terminal characteristic as shown in **Figure-(f)**. Let the voltage be $v_s = \cos \omega t$ V. What is the current i_D and v_D . (5)





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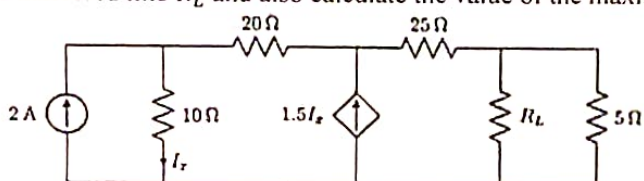
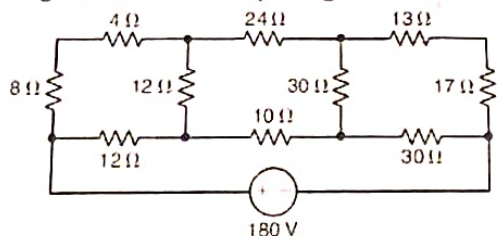
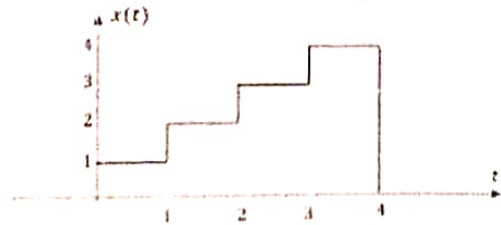
Email: sonepatiiit@gmail.com, website: www.iiitsonapat.ac.in

Mid Sem
Branch: CSE/IT/DSA

Subject: FEE (BEL-103)
M.M.-30 marks

Roll no
Time- 2 Hrs.

Note: All questions are compulsory.

Q-1: (a)	State and prove the Maximum Power Transfer Theorem and find the condition for maximum power transfer in a resistive network. Find the value of R_L in the below circuit, such that maximum power is delivered into R_L and also calculate the value of the maximum power. 	(5 marks)
(b)	Determine the current through 10Ω resistance by using star-delta transformation. 	(5 marks)
Q-2: (a)	Derive the expressions for the impedance and phase angle of both series and parallel RLC circuits. With the help of neat phasor diagrams, explain the circuit behavior under lagging, leading, and unity power factor conditions.	(5 marks)
(b)	Find the average value, RMS value, form factor and peak factor for the given stair signal. 	(5 marks)
Q-3: (a)	With a neat sketch, explain the B-H curve and describe how it is affected by material properties. Compare the analogy between electric and magnetic circuits.	(5 marks)
Q-4 (a)	A single-phase transformer has 2300/230 V, 50 Hz rating. The maximum flux in the core is 0.012 Wb. (a) Calculate the number of turns on the primary and secondary windings. (b) Find the EMF per turn. (c) If the transformer is operated at 60 Hz with the same maximum flux, what should be the new primary voltage?	(5 marks)

Roll No:

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B.Tech First Sem Mid Semester Examination, 03 Nov, 2025

Mid semester

Subject: Environmental Studies

Course Code: HUL107

Branch: CSC/IT/DSA

Maximum Marks. 30 marks

Time- 02 Hours.

Note: All questions are compulsory.

Q-1	Explain the different types of natural resources with examples and why is Conservation of natural resources important ?	(05 Marks)
Q-2	Explain the major threats to biodiversity with at least three examples ?	(05 Marks)
Q-3	Define environmental pollution? Mention its major types and give one example for each ?	(05 Marks)
Q-4	What are fossil fuels ? Why are they considered non-renewable resources?	(03 Marks)
Q-5	What is biodiversity hotspot? Name any two biodiversity hotspots in India ?	(03 Marks)
Q-6	Discuss the role of communities and traditional knowledge in the conservation of biodiversity?	(03 Marks)
Q-7	Discuss the effects of plastic pollution on the environment and wildlife?	(03 Marks)
Q-8	Write a short note on the role of the government and citizens in controlling environmental pollution?	(03 Marks)



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 Email: sonepatiiit@gmail.com, website: www.iiitsonepat.ac.in
 B.Tech.1st Sem Mid Semester Examination, 27th October, 2025.

Roll No.

Subject: Engineering Mathematics-I

Branch: CS/IT/DS

Max.Marks.-30

Subject Code: MAL101

Time- 2 Hours

Note: All questions are compulsory. (Scientific Calculator is not Allowed)

Q-1	Show that the matrix $A = \begin{bmatrix} 3 & 1 & -1 \\ -2 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix}$ is diagonalizable. Hence, find Matrix P such that $P^{-1}AP$ is a diagonal matrix. And also obtain the matrix $B = A^2 + 5A + 3I$.	3 Marks
Q-2	Reduce the matrix $A = \begin{bmatrix} 2 & 1 & 3 & 4 \\ 0 & 3 & 4 & 1 \\ 2 & 3 & 7 & 5 \\ 2 & 5 & 11 & 6 \end{bmatrix}$ to Normal Form and hence find its rank.	3 Marks
Q-3	Find the solution of system of linear equations, $x_1 + 6x_2 + 2x_3 = 9$, $-x_1 - 3x_2 - x_3 = 17$, $2x_1 + 12x_2 + 5x_3 = -4$ By Using LU decomposition Method.	3 Marks
Q-4	Find the Singular Value Decomposition of the Matrix $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix}$, that $A = U \Sigma V^T$.	3 Marks
Q-5	Expand $f(x) = e^{x \cos x}$ by Maclaurin's series upto three terms containing x^3 .	3 Marks
Q-6	Find the radius of curvature ' ρ ' of the curve $x^{2/3} + y^{2/3} = a^{2/3}$ at the points $x = a \cos^3 \theta$, $y = a \sin^3 \theta$.	3 Marks
Q-7	Trace the Cartesian Curve $y^2(a+x) = x^2(a-x)$, $a > 0$ Completely and hence sketch the shape of given curve including its tangents and Asymptotes.	3 Marks
Q-8	Show that the function $f(x,y) = \frac{x^2+y^2}{ x + y }$, when $(x,y) \neq (0,0)$ and equal to zero when $(x,y) = (0,0)$ is Continuous at $(0,0)$, but its Partial Derivatives f_x and f_y do not exist at $(0,0)$.	3 Marks
Q-9	If $\frac{x^2}{a^2+u} + \frac{y^2}{b^2+u} + \frac{z^2}{c^2+u} = 1$, then prove that $\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial u}{\partial z}\right)^2 = 2 \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z}\right)$.	3 Marks
Q-10	If $T(x, y, z) = (x^2 + y^2 + z^2)^{m/2}$, then find the value of m such that $(m \neq 0)$ which will make $\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = 0$.	3 Marks



Roll No:

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Email: sonapatiiit@gmail.com, website: www.iiitsonapat.ac.inB. Tech Ist Sem Mid Semester Examination, 29th October, 2025

Exam-Mid Sem

Subject: Applied Science

Course Code: BSL-104

Branch: CS+IT+DSA

Maximum Marks: -30 Marks

Time- 2 Hours.

Note: All questions are compulsory.

Q-1	a) Derive Schrödinger's time independent wave equation b) Explain the physical significance of wave function	(5 marks= 4+1)
Q-2	a) Describe the behavior of particles in a one-dimensional infinite potential well in terms of Eigen values and Eigen function. b) An electron is bounded in a one-dimensional infinite well having a width of 1×10^{-10} m. Find the energy values in the ground state and the first two excited states in terms of eV.	(5marks=3.5+1.5)
Q-3	a) What are the postulates of classical free electron theory? b) Derive an expression for electrical conductivity in a metal by using Drude & Lorentz theory c) Find relaxation time of conduction electron in metal if its resistivity is $1.54 \times 10^{-8} \Omega\text{-m}$ and it has 5.8×10^{28} conduction electron/ m^3 . Given $m = 9.1 \times 10^{-31}$ kg, $e = 1.6 \times 10^{-19}$ C	(2+3+2=7marks)
Q-4	a) Explain Heisenberg uncertainty principle b) The position of an electron in an atom is located within a distance of $0.1 \text{ \AA}$ using a microscope. What is the uncertainty in the momentum of the electron located in this way?	(2+1=3 marks)
Q-5	a) When two or more waves with slightly different velocity and frequency superimpose on each other, Derive the expression for their group velocity b) Find the phase velocity and group velocity of an electron whose de Broglie wavelength is $1.2 \text{ \AA}$	(3+2 marks)
Q-6	a) Consider the light beam coming from a 1mW He-Ne ($\lambda=633$ nm) laser source. How many photons are there in 1meter length of the beam? b) Let $\psi = A e^{-x^2/a^2} e^{ipx/\hbar}$ Calculate $X P_x \psi$ and $P_x X \psi$, where X and P_x are position and momentum operators.	(3+2 marks)

$$9.1 \times 10^{-31} \text{ kg}$$



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Mid Semester Examination

Subject Name: Computer Programming

Subject Code: UCS001

Roll no

Branch: CSE/IT/DSA

Semester 1st

M.M.-30 marks

Time- 120 min.

Note: All questions are compulsory. Assume all missing data.

Q.1	What will be the output of the following programs. a. <pre>#include <stdio.h> int main() { int i = 3; printf("%d %d %d\n", i, ++i, i++); return 0; }</pre> b. <pre>#include <stdio.h> int main() { int a = -3, b = 3; double z = (a < 0 ? a * 1.0 : b) + (b > 0 ? 0.5 : 1); printf("%.1f\n", z); return 0; }</pre> c. <pre>#include <stdio.h> int main() { int a = 10; if (a > 5) if (a < 15) printf("Inside"); else printf("Outside"); printf(" Done"); return 0; }</pre> d. <pre>#include <stdio.h> int main() { int a[] = {1, 3, 5, 7, 9}; int i = 1; int s = a[i] + i[a] + *(a + 2); printf("%d\n", s); return 0; }</pre> e. <pre>#include <stdio.h> #define MAX(a,b) ((a)>(b) ? (a):(b)) int main(void){ int x=1, y=1; MAX(x++, y++); printf("%d %d\n", x, y); }</pre>	(5 marks)
Q.2	a. Write a function that returns 1 if an integer is a perfect number, else 0. b. With a neat flowchart outline the steps to compute the GCD of two numbers.	(5 marks)
Q.3	a. Explain how break, continue, and goto statements affect the flow of control in a C program. b. Write a c program to swap two numbers using function.	(5 marks)
Q.4	Explain the difference between 1D and 2D arrays in C. How are they declared and initialized? Write a C program to input and display elements of a 3x2 array using nested loops and display the sum of its diagonal elements.	(5 marks)
Q.5	Write a C program to find the second largest number among four numbers using if-else only (no loops or arrays).	(5 marks)
Q.6	Write short note on the followings with an example. a. Sizeof Operator b. Switch statement c. Recursion in C d. Nested if else e. Do while loop	(5 marks)

End